

Ch. Neha Singha

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SKILLS

- **Programming Languages:** Embedded C, Java, HTML, CSS
- **Microcontrollers & Hardware:** ESP32, Arduino Uno, 8051 Microcontroller
- **Embedded Dev & Simulation Tools:** Keil uVision, Proteus, Wokwi, Arduino IDE
- **Software & Data Tools:** MATLAB, MySQL, Power BI, Microsoft Suite
- **Soft Skills:** Strategic Thinking & Problem Solving, Teamwork & Interpersonal Skills, Time Management, Analytical Skills

INTERNSHIP

Gurudev Projects Constructions Pvt. Ltd.

Jun – Jul 2025

Data Analyst Intern

- Cleaned, organized, and structured raw construction-project datasets using Excel and Power BI, producing weekly monitoring reports that improved reporting turnaround time.
- Streamlined documentation and record-keeping workflows, improving data accuracy and reducing record-retrieval time for management decision-making.

PROJECTS

Sign-to-Speech Communication Device (ESP32)

May 2026

- Designed and developed a wearable sign-to-speech communication device using an ESP32 and an MPU6050 motion sensor, achieving real-time recognition of 10+ hand gestures.
- Programmed embedded C firmware to convert recognized gestures into audio output in under 1 second, cutting communication delay for users with speech or hearing impairments.
- **Tech Stack:** Embedded C, ESP32, MPU6050, Edge Impulse

Smart Safety Pendant (Arduino)

Apr 2026

- Engineered a wearable emergency-alert system with gesture-based (tap/shake) activation, enabling discreet SOS triggering for silent emergencies.
- Integrated Arduino Uno, an accelerometer, a touch sensor, and a GSM module to send SMS alerts.
- **Tech Stack:** Embedded C, Arduino Uno, GSM Module

Smart Gesture-Controlled Robotic Car (ESP32)

Dec 2025

- Designed a gesture-controlled robotic car using ESP32 and a motion sensor, achieving accurate recognition of directional gestures (forward, reverse, turn).
- Developed embedded C firmware for wireless control, reducing command-response latency for smoother real-time operation.
- **Tech Stack:** Embedded C, ESP32, IoT

CERTIFICATES

- Embedded Systems with AVR ATMEGA32 Microcontroller | [Udemy](#) Feb 2026
- Power BI | [Udemy](#) Feb 2026
- Introduction to Front End Development | [Coursera](#) Jan 2026
- Database Management System | [NPTEL](#) Sep 2025

EDUCATION

Lovely Professional University | CGPA: 7.48

Punjab, India

Bachelor of Technology - Electronics and Communication Engineering

Aug 2023 – Present

Kendriya Vidyalaya Narengi | Percentage: 87.4

Assam, India

Intermediate (PCMB)

Apr 2020 – Mar 2022

Kendriya Vidyalaya Narengi | Percentage: 93.6

Assam, India

Matriculation

Apr 2019 – Mar 2020